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Fair Split Bill A Meal Cost Sharing System Based on Actual Consumption



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Five friends—Aurelius, Belsyazar, Cassian, Dora, and Tom—meet at a famous noodle restaurant after class. To protect their privacy, these names are fictional. Because everyone orders and consumes different amounts, dividing the total bill equally may not be fair.

- Aurelius orders 2 bowls of noodles and 1 drink.
- Belsyazar orders 1 bowl of noodles, 1 drink, and eats 1 piece of dimsum.
- Cassian orders 1 piece of dimsum and has 1 drink.
- Dorian orders 1 bowl of noodles, 1 drink, and eats 3 pieces of dimsum.
- Tom eats 1 bowl of noodles, 1 drink, and eats 2 pieces of dimsum.

The menu prices are:

- Noodle: Rp10.000 per bowl
- Dimsum: Rp9.000 per portion
- Drink: Rp5.000 per cup

Person	Amount paid
Aurelius	Rp19.200
Belsyazar	Rp19.200
Cassian	Rp19.200
Dorian	Rp19.200
Tom	Rp19.200
Total	Rp96.000

Person	Noodles	Dimsum pieces	Drink	Fair amount
Aurelius	Rp20.000	Rp0	Rp5.000	Rp25.000
Belsyazar	Rp10.000	Rp3.000	Rp5.000	Rp18.000
Cassian	Rp0	Rp3.000	Rp5.000	Rp8.000
Dorian	Rp10.000	Rp9.000	Rp5.000	Rp24.000
Tom	Rp10.000	Rp6.000	Rp5.000	Rp21.000
Total	Rp50.000	Rp21.000	Rp25.000	Rp96.000

Person	Equal	Fair	Diff
Aurelius	Rp19.200	Rp25.000	-Rp5.800
Belsyazar	Rp19.200	Rp18.000	+Rp1.200
Cassian	Rp19.200	Rp8.000	+Rp11.200
Dorian	Rp19.200	Rp24.000	-Rp4.800
Tom	Rp19.200	Rp21.000	-Rp1.800

Why Equal Split Is Unfair?

- People who eat less may pay more than they should
- People who eat more may pay less than they should
- Shared food is often not divided according to actual portions
- Drinks and extra items are often not recorded accurately

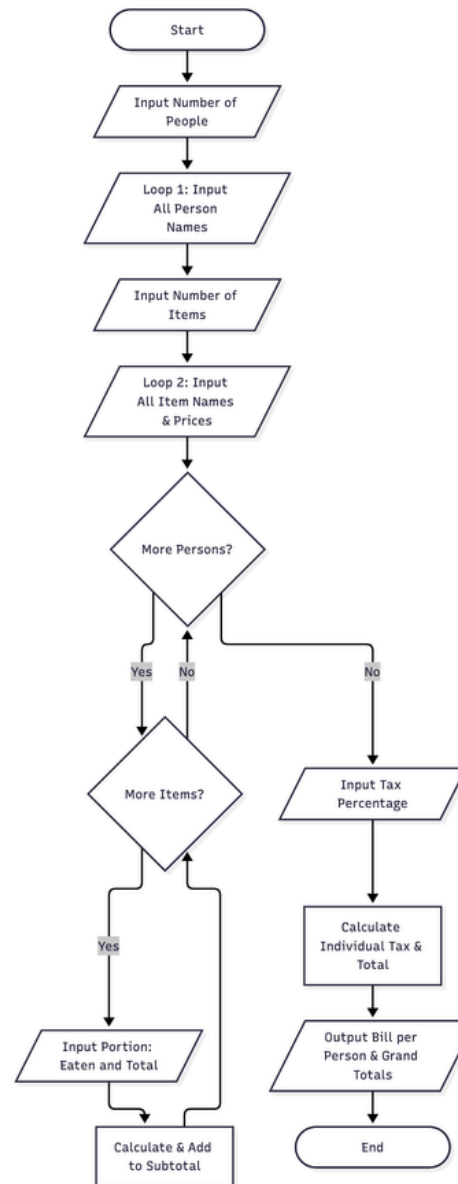
Proposed Concept

1. Input each person's name
2. Input the list of food and drinks
3. Identify who consumed each item
4. If an item is shared, divide it by portion or quantity
5. Calculate each person's total based on actual consumption
6. Tax or service charge can be divided proportionally

Advantages of Using C++

- High performance for real-time calculation - Fast calculations and instant results.
- Offline operation without requiring network access - Works without Wi-Fi or mobile data.
- Portable across multiple desktop operating systems - Can run on Windows, Linux, and macOS.
- supports object-oriented programming - Uses classes and objects for organized code.
- Input Validation - Detects invalid inputs and prevents errors.
- File handling - Saves bills, transactions, and receipts for later use.

Flowchart:



Algoritmn:

1. Input the total item amount
2. Divide the item per person
3. Calculate in this formula:

$$\text{Personal Subtotal} = \sum \left(\text{Item Price} \times \frac{\text{Person's Portion}}{\text{Total Portions}} \right)$$

4. Toggle Tax or No
5. Output amount for each person

Conclusion:

The Fair Split Bill system addresses the inherent inequities of traditional bill-splitting by replacing arbitrary division with a precise, data-driven algorithm. By leveraging the robust capabilities of C++, developed a high-performance, offline-capable solution that meticulously calculates individual consumption, accurately apportions shared items, and distributes taxes proportionally. Ultimately, this project not only demonstrates the practical application of object-oriented programming and reliable file handling, but it also delivers a highly equitable tool that definitively resolves the everyday financial friction of shared dining expenses.

REFERENCES:

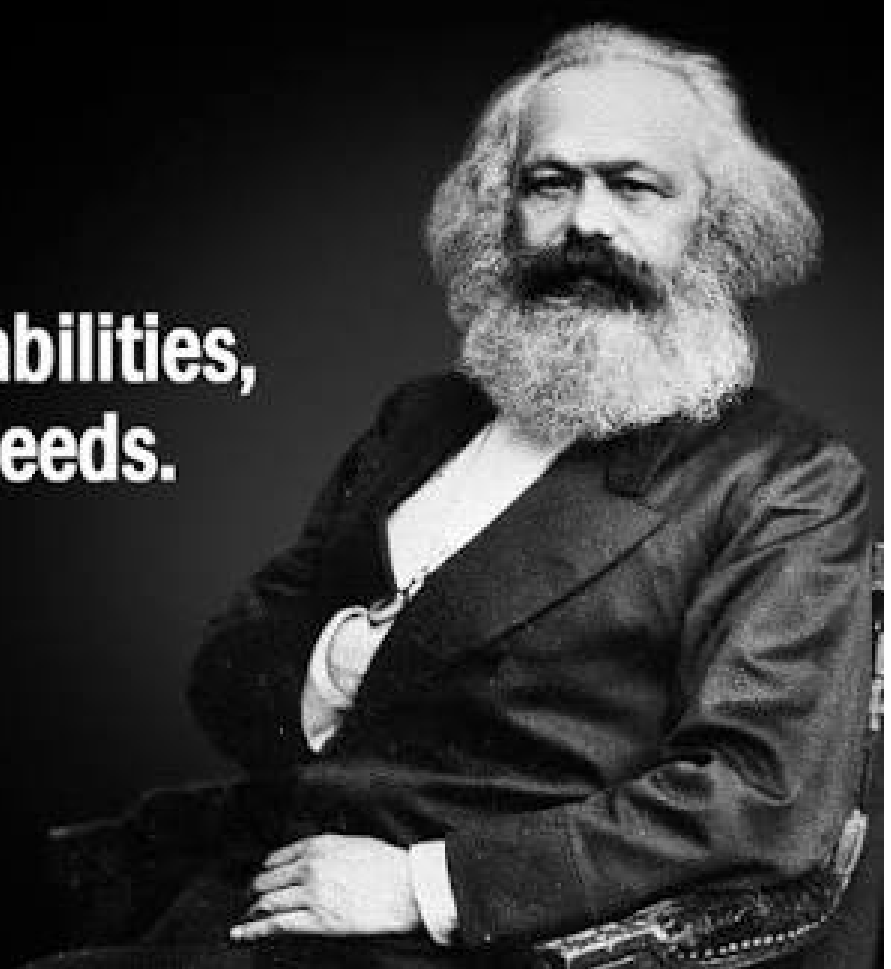
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From each according to his **abilities**,
to each according to his **needs**.

– *Karl Marx*

AZ QUOTES





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THANK YOU!



*Merakyat, Mandiri, Berkelanjutan
Inclusive, Self-Reliant, Sustainable*

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